

②

$$Q = 3 \times 10^{-6}$$

$$R = 3 \text{ cm}$$

case (i)

$$f = \frac{Q_1 Q_2}{4\pi\epsilon R^2}$$

$$f = \frac{Q}{4\pi\epsilon R^2} \quad D = \epsilon E$$

$$Q = 4 \times 10^{-6} \text{ C}$$

$$R = 3 \text{ cm}$$

(i) electric field intensity

$$E = \frac{Q}{4\pi\epsilon R^2}$$

$$= \frac{3 \times 10^{-6}}{4 \times 3.14 \times 8.85 \times 10^{-12} (3 \times 10^{-2})^2}$$

$$= 1.8766 \times 10^4$$

$$= 3 \times 10^{-6}$$

$$1.000856 \times 10^4$$

$$= 2.9974 \times 10^{15} \text{ V/m}$$

2) electric flux density

$$D = \epsilon E$$

$$= (8.85 \times 10^{-12}) (2.9974 \times 10^{15})$$

$$= 2.65389 \text{ C/m}^2 \Rightarrow 0.002653$$

case (ii)

electric flux density

$$D = \epsilon E$$

$$= (8.85 \times 10^{-12}) (8.445 \times 10^{18})$$

$$= 7.50332 \text{ C/m}^2$$

Electrical field intensity

$$\begin{aligned} E &= \frac{Q}{4\pi\epsilon r^2} \\ &= \frac{3 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} (3)^2} \\ &= \frac{3 \times 10^{-6}}{3.54 \times 10^{-11}} \\ &= 8.4745 \times 10^{-18} \text{ V/m} \end{aligned}$$

case 3:-

1) electric field intensity $E = \frac{Q}{4\pi\epsilon r^2}$

$$\begin{aligned} &= \frac{3 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times (3)^2} \\ &= \frac{3 \times 10^{-6}}{3.186 \times 10^{-10}} \\ &= 9.41619 \times 10^{-17} \text{ V/m} \end{aligned}$$

Electric field intensity

$$\begin{aligned} E \cdot \epsilon &= \epsilon E \\ &= (9.41619 \times 10^{-17}) \times (8.85 \times 10^{-12}) \text{ V/m} \\ &= 8.33332 \text{ C/m}^2 \end{aligned}$$